



The Art of ECRR

by BossCat — IONA User Manual

"Speed is born from discipline. Discipline is born from rules that end cleanly."

I. The Four Letters (Tetragram)

ECRR = Evidence → Contain → Rollback → Report. When in doubt, **ECRR and terminate.**

- **Evidence:** Capture facts (logs, state, diffs).
 - **Contain:** Stop blast radius (halt writes, freeze lane).
 - **Rollback:** Return the repo and job state to last-known-good.
 - **Report:** Emit a concise incident record to BossCat and exit.
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II. Roles, Lanes, and Budgets

- **A (Actor):** the only writer; executes the job.
- **B (Balancer):** reads, verifies, times, and reports; never writes to the repo.
- **BossCat:** command channel / orchestrator.
- **Lane:** short-lived branch or workspace assigned to a single job.
- **Budgets** (hard): ≤ 2 jobs, ≤ 10 files, ≤ 200 LOC touched per job. *Outside the lane or over budget is forbidden.*

Repo contracts (created by A only):

- `.agent/JOB.lock` — mutex for the active job.
 - `.agent/EVIDENCE.log` — JSONL events (append-only).
 - `.agent/PLAN.md` — one-page intent & scope (optional but recommended).
 - Kill-switch (human ops/infra): `.agent/LOCK` — if present, all bots stop.
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III. The Prime Rules

Rule #1 — Two Make the Strike

Auto-bots operate in pairs. **A** implements the task. **B** monitors A's output and reports to BossCat.

- If **A** cannot locate **B** (heartbeat missing or channel down): **ECRR and terminate.**
- If **B** cannot locate **A** (no lock, no logs, or stalled TTL): **ECRR and terminate.**

"A single sword cuts crooked. A paired guard keeps the edge straight."

Checklist (B)

1. Watch `.agent/JOB.lock` heartbeat (mtime tick ≤ 60 s).
 2. Tail `.agent/EVIDENCE.log`; enforce TTL & budgets.
 3. On anomaly: page BossCat, then require ECRR.
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Rule #2 — Single-Writer, Lane-Locked

A is the only writer; B never writes.

- Before A edits, acquire mutex: create `.agent/JOB.lock`.
If lock already exists, **ECRR and terminate**.
- A touches only files inside the **active lane**.
- Respect hard budgets: ≤ 2 jobs, ≤ 10 files, ≤ 200 LOC.

"One pen, one path, one purpose."

Checklist (A)

1. `if [ -e .agent/LOCK ]; then ECRR; exit 1; fi`
 2. `if [ -e .agent/JOB.lock ]; then ECRR; exit 1; else touch .agent/JOB.lock; fi`
 3. Emit PLAN (`.agent/PLAN.md`), then begin.
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Rule #3 — No-Human / No-Conflict Preflight

- Abort instantly if `.agent/LOCK` exists.
- Abort if the lane is not fast-forward clean against its base (no conflicts, no pending human changes).
- Tools, tokens, and secrets must be present and scoped; missing pre-reqs → **ECRR**.

"Enter only quiet rooms. Leave no prints but your own."

Checklist (A)

- Verify: clean worktree, sync base, deterministic env, secrets scope OK.
 - Dry-run diff $\leq$ budgets; else **ECRR**.
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Rule #4 — Bounded Retry, or Evidence and Stop

Wrap every action with crash-safe retry: ≤ 3 retries with exponential backoff; **job TTL** applies.

- On final failure: **rollback**, then **ECRR and terminate**.

- B enforces that repeated flaps do not exceed TTL.

"Try thrice; the fourth time is folly."

IV. The Campaign Rules (Operational Doctrine)

Rule #5 — Declare Intent Before You Move

Before the first write, A commits a plan of attack.

- Create/append `.agent/PLAN.md` (≤ 150 words): *goal, scope, files, tests*.
- Log a `plan` event in `.agent/EVIDENCE.log`.

"Announce the hill; then take it."

Rule #6 — Evidence-First Telemetry

Every significant step writes one JSON line to `.agent/EVIDENCE.log`.

Event schema (one line each):

```
{ "t": "ISO8601", "who": "A|B", "type": "plan|preflight|lock|edit|test|rollback|  
report|exit",  
  "lane": "<branch>", "files_touched": 3, "loc_delta": 72, "msg": "<short>" }
```

B never edits this file; B only reads it.

"What is not written did not happen."

Rule #7 — CI Gate: Changed-Paths Only

A must run **smoke tests** that cover only changed paths.

- If tests fail: rollback → **ECRR**.
- Long suites are forbidden; keep latency low.

"Test the bridge you cross, not the river you leave behind."

Rule #8 — Review Without Writing

B reviews artifacts and diffs; B posts **out-of-repo** status to BossCat (e.g., API/chat).

- If evidence insufficient (no plan, missing tests, over budget): command **ECRR**.
- B may not push commits, resolve conflicts, or edit files.

"Eyes open; hands off."

Rule #9 — Merge Is Not a Bot's Honor

Bots do **not** merge to trunk. A opens the lane; integration is external (BossCat/human ops).

- Rebase lane to base on start and before exit; if conflict appears → **ECRR**.

"Win the skirmish; leave the treaty to others."

Rule #10 — Secrets and Boundaries

- No new external endpoints, no credential sprawl, no writing outside lane paths.
- Large binaries (> 10 MB) and unvetted dependencies are forbidden.
- Any secret exposure → immediate **Contain** → **Rollback** → **Report**.

"Guard your keys; narrow your doors."

Rule #11 — Artifacts Are Ephemeral

- Temporary build outputs live under `.agent/tmp/` and are deleted on success.
- Persistent artifacts (if any) are checksummed and listed in `.agent/PLAN.md`.

"Carry light; leave nothing shining."

Rule #12 — Latency Comes from Small Cuts

Prefer **many small edits** over one broad change.

- Keep diffs surgical; pre-warm caches if available; stream evidence early.

"Thin blades enter first."

Rule #13 — Health, Heartbeat, and TTL

- A updates the mtime of `.agent/JOB.lock` every ≤ 60 s.

- **Default TTL** per job: 15 minutes (tune per org).
- **Stall** = no heartbeat for 2 intervals → B commands **ECRR**.

"A quiet drum is a fallen unit."

Rule #14 — Rollback Is a First-Class Action

- A maintains a revert plan for every edit (stash/patch/branch).
- On failure: restore last-known-good, then log `rollback`.

"Retreat cleanly so you may return quickly."

Rule #15 — Exit Codes and Colors (NATO-style)

- **GREEN (0)**: success; lane rebased; evidence complete.
- **AMBER (10)**: soft stop (budget/TTL reached); no writes pending; report.
- **RED (20)**: hard failure; ECRR executed.
- **BLACK (30)**: kill-switch or policy breach; ECRR executed, escalation required.

"Name the state; command the response."

Rule #16 — Manual Changes Are Sacred

If a human touches the lane during a bot job (detected via author/committer):

- Immediate **Contain** (freeze), then **ECRR** and terminate.

"Where humans walk, machines kneel."

V. Communications (Brevity Codes)

Call-signs: `A`, `B`, `BossCat`.

B → BossCat status template (out-of-repo):

```
IONA/BALANCER STATUS
Lane: <branch> TTL: <mm:ss> Budget: <files>/<loc>
State: GREEN|AMBER|RED|BLACK
Last event: <t> <type> "<msg>"
Action: continue|hold|ECRR
```

VI. Minimal CI/CD Wire-Up (Fast, Not Fancy)

1. **Trigger:** BossCat assigns lane & budgets; pairs A/B.
 2. **Preflight** (A): No kill-switch, acquire lock, declare plan, log `preflight`.
 3. **Execute** (A): Edit within budgets; log `edit`.
 4. **Smoke** (A): Run changed-paths tests; log `test`.
 5. **Verify** (B): Read evidence & diffs; if OK, advise **GREEN** to BossCat.
 6. **Exit** (A): Rebase-check only; never merge; drop lock; log `exit`.
 7. **On Any Breach: ECRR and terminate.**
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VII. ECRR Pocket Card (do this verbatim)

When any rule demands “ECRR and terminate,” A performs:

1. Evidence

2. Append to `.agent/EVIDENCE.log` the final `report` with reason, files, and loc.
3. Save `git diff --no-color` patch to `.agent/last.diff` (if present).

4. Contain

5. Stop all writes; preserve `.agent` as the only added files.
6. Freeze lane (no more retries).

7. Rollback

8. Revert worktree to base tip; delete `.agent/tmp/`.
9. Leave `.agent/JOB.lock` in place until Report succeeds.

10. Report

11. B posts status **RED/BLACK** to BossCat with a 3-line summary.
12. A removes `.agent/JOB.lock` and exits with the proper code.

No improvisation. No partial cleans. End cleanly.

VIII. Glossary (quick)

- **Lane:** short-lived branch/workspace for a single job.
- **Budget:** limits on jobs, files, and LOC.

- **TTL:** job time-to-live.
 - **ECRR:** Evidence, Contain, Rollback, Report (the “four letters”).
 - **Terminate:** clean exit with state restored and status reported.
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